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ORIGINAL RESEARCH ARTICLE

Dark-adapted threshold and electroretinogram for diagnosis of Usher syndrome

Lucia Ambrosio · Ronald M. Hansen · Anne Moskowitz · Andrea Oza · Devon Barrett · Juliana Manganella · Genevieve Medina · Kosuke Kawai · Anne B. Fulton · Margaret Kenna

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Abstract

Purpose To determine the utility of ophthalmology evaluation, dark-adapted threshold, and full-field electroretinogram for early detection of Usher syndrome in young patients with bilateral sensorineural hearing loss.

Methods We identified 39 patients with secure genetic diagnoses of Usher Syndrome. Visual acuity, spherical equivalent, fundus appearance, dark-adapted threshold, and full-field electroretinogram results were summarized and compared to those in a group of healthy controls with normal hearing. In those Usher patients with repeated measures, regression analysis was done to evaluate for change in visual acuity and dark-adapted threshold with age. Spherical equivalent and full-field electroretinogram responses from dark- and light-adapted eyes were evaluated as a function of age.

Results The majority of initial visual acuity and spherical equivalent results were within normal limits for age. Visual acuity and dark-adapted threshold worsened significantly with age in Usher type 1 but not in Usher type 2. At initial test, full-field electroretinogram responses from dark- and light-adapted eyes were abnormal in 53% of patients. Remarkably, nearly half of our patients (17% of Usher type 1 and 30% of Usher type 2) would have been missed by tests of retinal function alone if evaluated before age 10.

Conclusions Although there is an association of abnormal dark-adapted threshold and full-field electroretinogram at young ages in Usher patients, it appears that a small but important proportion of patients would not be detected by tests of retinal

L. Ambrosio (✉) · R. M. Hansen · A. Moskowitz · A. B. Fulton
Department of Ophthalmology, Boston Children's Hospital, 300 Longwood Avenue, Boston, MA 02115, USA
e-mail: Lucia.ambrosio@childrens.harvard.edu

L. Ambrosio · R. M. Hansen · A. Moskowitz · A. B. Fulton
Department of Ophthalmology, Harvard Medical School, 25 Shattuck Street, Boston, MA 02115, USA

A. Oza
Laboratory for Molecular Medicine, Partners HealthCare Personalized Medicine, 65 Landsdowne Street, Cambridge, MA 02139, USA

D. Barrett · J. Manganella · G. Medina · K. Kawai · M. Kenna
Department of Otolaryngology and Communication Enhancement, Boston Children's Hospital, 300 Longwood Avenue, Boston, MA 02115, USA

K. Kawai · M. Kenna
Department of Otolaryngology, Harvard Medical School, 25 Shattuck Street, Boston, MA 02115, USA

function alone. Thus, genetic testing is needed to secure a diagnosis of Usher syndrome.

Keywords Usher syndrome · Full-field electroretinogram · Dark-adapted visual threshold · Sensorineural hearing loss

Introduction

Usher syndrome (USH) is the most common heritable cause of combined vision and hearing impairment, affecting between 3.3/100,000 and 6.2/100,000 in most populations [1–4]. Inheritance is autosomal recessive, involving mutations in any of 9 confirmed genes, with several others under consideration [5]. Typically, USH presents with hearing loss before visual loss. Patients are categorized by clinical subtypes, with USH type 1 (USH1) having severe to profound congenital bilateral sensorineural hearing loss (SNHL), symptomatic retinal degeneration starting in late childhood or adolescence, and vestibular dysfunction [6–10]. Those with USH type 2 (USH2) typically exhibit moderate-to-severe congenital bilateral SNHL, retinal degeneration starting during adolescence or early adulthood, and no vestibular symptoms [7, 11]. USH type 3 (USH3), the rarest type, causes progressive SNHL and variable retinal and vestibular dysfunction [11, 12].

Early and accurate diagnosis of USH remains challenging. Because of the heterogeneity of hearing loss in USH, diagnosis based on audiograms alone is impossible [13]. Standard ophthalmic examination is part of the comprehensive clinical assessment, even though, in some cases, the child may have no symptoms of retinal degeneration and no retinal signs detectable by ophthalmoscopy [14].

In one study among children with congenital SNHL who tested negative for the most common form of non-syndromic hearing loss (Connexin 26 protein, encoded by the *GJB2* gene), the prevalence of ≥ 1 Usher mutation was reported as 11.3% [15]. In a second study, among 227 deaf children who appeared to be non-syndromic at presentation, 5 (2.2%) had biallelic USH1 mutations [16]. In a third recently published study, 18/184 children with clinically non-syndromic SNHL were found to have biallelic pathogenic/likely pathogenic (9) or variants of unknown significant (9) in genes associated with USH. Of these 18, 14 had ophthalmology evaluation, and 4/14 had previously unsuspected retinal abnormalities [17]. Abnormal full-field electroretinogram (ERG) responses and elevated dark-adapted threshold (DAT) have been reported in patients with USH [12, 14, 18–22]. Early genetically confirmed diagnosis of USH based on abnormal ERG and DAT testing before vision loss manifests would inform the child's management and provides information to guide prognosis and educational and therapeutic choices.

Methods

Patients

The patients were seen at Boston Children's Hospital in the Departments of Ophthalmology and/or Otolaryngology from February 1, 1994, through August 1, 2019. We identified 39 pediatric patients with bilateral SNHL and biallelic USH mutations (including USH1, USH2, USH3) and analyzed the results of their full-field ERG and DAT tests as well as their visual acuity (VA), refraction, and fundus appearance and compared their ophthalmological findings with two groups of age-similar controls: those having non-syndromic biallelic *GJB2*-related SNHL and healthy normal hearing controls.

The DAT results from the USH patients were compared with the thresholds of 13 children with non-syndromic sensorineural hearing loss due to biallelic pathogenic *GJB2* mutations and with 26 healthy young adults [22]. The *GJB2* controls, aged 1.5 to 14.4 (median 10) years at time of testing, were in good general health; thorough full eye examination (VA, motility, alignment, pupillary reactions, biomicroscopy of anterior and posterior segments, ocular pressure, cycloplegic streak retinoscopy, and indirect ophthalmoscopy) showed that they had normal eyes.

The ERG results from the Usher patients were compared with 56 healthy controls [23, 24].

Diagnostic genetic testing was performed in a Clinical Laboratory Improvement Amendments (CLIA)-approved laboratory; diagnosis methods included Sanger sequencing, re-sequencing microarray, and next-generation sequencing.

Audiometry was performed using one or more developmentally appropriate behavioral techniques.

Pure-tone average was calculated by taking the average of responses at 500, 1000, 2000, and 4000 Hz for each ear and categorized into six severity levels, normal to profound [25]. Patients with no response at any of these frequencies were categorized as profound [26].

Eye examination

Visual acuity was assessed using age-appropriate tests. Motility, alignment, and pupillary reactions to light were evaluated. Cycloplegic streak retinoscopy was used to determine refractive error. The structures of the anterior and posterior segments were examined using slit lamp biomicroscopy and indirect ophthalmoscopy for signs of retinal degeneration (cells in the retrolental space, arteriolar narrowing, abnormal pigment in the macula and/or extramacular fundus, and pallor of the optic nerve head).

Dark-adapted threshold

The dark-adapted threshold (DAT) is a test of retinal sensitivity. It is applicable at all ages, from early infancy onward. Thus, it suits our purpose of making longitudinal measurements starting even in infancy.

Before testing, the patient is dilated and dark-adapted for 30 min. Infants and children are accompanied by a parent. A two-alternative, spatial, forced-choice psychophysical procedure in conjunction with a staircase method is used to determine threshold [22]. Stimuli are 10° diameter, 50 ms duration, blue ($\lambda < 510$ nm) spots presented 20° to the left or right of a small central fixation target; all are shown on a rear projection screen. On each trial, the patient reports stimulus position (right or left) and receives feedback from the tester (correct or incorrect). For infants and young children, an adult observer, equipped with an IR viewer and unaware of stimulus location, reports stimulus location, right or left, based on the patient's looking responses. In every trial, the observer receives feedback from the tester. For healthy, mature control participants, the average DAT is −5.6 (0.12) log cd s/m^2. In a healthy mature eye with an 8-mm pupil, this produces -3.9 log scot td sec retinal illuminance [22].

Electroretinogram (ERG)

Responses to brief, full-field stimuli are recorded using previously described methods [27, 28]. Before testing, the patient's pupils are dilated and the patient is dark-adapted for 30 min. Then, under dim red light, following instillation of proparacaine, a bipolar Burian Allen electrode is placed on each eye and a ground electrode on the skin over one mastoid. Responses are amplified (gain = 2,000; band pass 0.5 to 1,000 Hz), displayed, averaged, and stored. Any responses contaminated by artifacts are rejected.

The stimuli included are similar to those recommended in the ISCEV standard [29]. For the purposes of this report, the responses of the dark-adapted retina to 0.01 and 0.23 cd s/m^2 stimuli are evaluated; 4 to 64 responses are averaged. In the light-adapted condition, responses to 2.25 cd s/m^2 flashes in the presence of a steady 30 cd/m^2 background are evaluated; 4 to 64 responses are averaged; and responses to 30.2 Hz flickering light are also evaluated; 4 to 256 responses are averaged. The b-wave is selected because, among these patients with retinal degeneration, more have detectable b-waves than a-waves. Only those responses > 2.1 μV are considered for analysis [30]. When it was possible, DAT testing and ERG testing were performed during the same session/day. Eighteen of thirty-eight patients were tested under anesthesia in coordination with cochlear implant surgery or auditory brainstem response testing. One patient with USH2 (#32 in Table 1) had no ERG test.

Data analysis

Visual acuity, spherical equivalent, and DAT were examined as a function of age and compared with the 95% prediction limits of normal for age [31–34]. For the longitudinal analysis, we employed a mixed-effects model that captures the collection of individual trajectories over time by incorporating random-subject effects. The linear mixed-effects model with random intercept and slope terms was used to account for within-subject correlations. We included as main effects age at test, type of USH, and the interaction between age and type of USH. All analyses were conducted using SAS version 9.4 (SAS Institute, Cary, NC) and SPSS version 20 and R statistical software. Data from the single patient having USH3 was not included in this analysis.

Table 1 Genetic diagnosis and ophthalmic characteristics by study patients

Pt #	USH type	Genetic diagnosis		First ophthalmic examination				
		Mutation 1	Mutation 2	Age (years)	Disc	Vessels	Fundus	Macula
USH 1								
1	MYO7A	c.640G > A; p.G214R	c.5573 T > C;p.L1858P	0.1	Normal	Normal	Normal	Normal
2	MYO7A	c.459_460delTG;p.A154fs‡	c.1583 T > G; p.L528R§	0.2	Normal	Normal	Normal	Normal
3	MYO7A	c.5101C > T; p.R1701*	c.2094 + 1G > A§	0.2	Normal	Normal	Normal	Normal
4	MYO7A	c.2904G > T;p.E968D	c.3532delC; p.Q1178fs§	0.2	Normal	Attenuated	Pigmentation	Altered pigment
5	MYO7A	c.496delG;p.E166fs	c.3327delC; p.H1109fs§	0.4	Normal	Normal	Normal	Normal
6	MYO7A	c.5886_5888delCTT; p.F1963del	c.5824G > T;p.G1942*§	0.4	Normal	Normal	Normal	Normal
7	MYO7A	c. 999 T > G;p.Y333*	c.4544_4551delinsCA; p.E1515_M1517delinsA	0.4	Normal	Normal	Normal	Normal
8	MYO7A	c.5886_5888delCTT;p.F1963del	c.5824G > T;p.G1942*	0.5	Normal	Normal	Normal	Normal
9	MYO7A	c.3591_3592delCT; p.C1198fs	c.3591_3592delCT; p.C1198fs	0.9	Normal	Attenuated	Pigmentation	Altered pigment
10	MYO7A	c.77C > A; p.A26E	c.3979G > A;p.E1327K	0.9	Normal	Normal	Normal	Normal
11	MYO7A	c.397_398insC; p.H133fs	c.5648G > A;p.R1883Q	1.1	Normal	Normal	Normal	Normal
12	MYO7A	c.5824G > T; p.G1942*	c.3829del; p.A1277Qfs*122‡	5.6	Normal	Attenuated	Pigmentation	Normal
13	MYO7A	c.2094 + 1G > C§	c.397C > T;p.H133Y	6.5	Normal	Attenuated	Pigmentation	Altered pigment
14	MYO7A	c.5824G > T; p.G1942*	c.3829del; p.A1277Qfs*122‡	8.8	Normal	Attenuated	Pigmentation	Normal
15	MYO7A	c.2005C > T;p.R669*	c.2005C > T;p.R669*	9.5	Normal	Attenuated	Pigmentation	Altered pigment
16	MYO7A	c.3764delA; p.K1255fs	c.722G > A;p.R241H	9.9	Normal	Attenuated	Pigmentation	Altered pigment
17	MYO7A	c.640G > A;p.G214R	c.587 T > C;p.L196§	12.7	Mild pallor	Attenuated	Pigmentation	Altered pigment
18	MYO7A	c.999 T > G;p.Y333*	c.999 T > G;p.Y333*	12.7	Normal	Attenuated	Pigmentation	Altered pigment
19	MYO7A	c.3591_3592delCT; p.C1198fs	c.3591_3592delCT; p.C1198fs	12.8	Normal	Attenuated	Pigmentation	Altered pigment
20	MYO7A	c.1690 + 1G > A	c.3544_3545insCA; p.N1182fs§	16.6	Mild pallor	Attenuated	Pigmentation	Altered pigment
21	MYO7A	c.3591_3592delCT; p.C1198fs	c.3591_3592delCT; p.C1198fs	17.4	Normal	Attenuated	Pigmentation	Altered pigment
22	MYO7A	c.2904G > A; p.E968E§	c.314 T > G;p.V105G	19.0	Mild pallor	Attenuated	Pigmentation	Normal
23	MYO7A	c.3591_3592delCT; p.C1198fs	c.3591_3592delCT; p.C1198fs	20.7	Normal	Attenuated	Pigmentation	Altered pigment
24	PCDH15	c.1997 + 1G > A§	c.1997 + 1G > A§	1.1	Normal	Normal	Normal	Normal
25	CDH23	c.7908C > G; p.Y2636*	c.7908C > G;p.Y2636*	3.1	Pink	Mildly attenuated	Pigmentation	Altered Pigment
26	CDH23	c.5272C > T; p.Q1758*‖	c.5712 + 1G > A‖	5.0	Normal	Normal	pigmentation	Normal
USH 2								
27	USH2A	c.2299delG; p.E767fs	c.6795_6797delATA; p.E2265_Y2266delinsD	1.2	Normal	Normal	Normal	Normal
28	USH2A	c.8981G > A; p.W2994*	c.9304C > T;p.Q3102*§	1.4	Normal	Normal	Normal	Normal
29	USH2A	c.1143 + 1G > A§	c.10190_10191delAA; p.K3397fs	1.4	Normal	Normal	Normal	Normal
30	USH2A	c.4338_4339delCT; p.C1447fs	c.11411delC; p.Pro3804LeufsX13)	3.1	Normal	Normal	Normal	Normal
31	USH2A	c.2299delG; p.E767fs	c.6224G > A;p.W2075*	4.9	Normal	Attenuated	Pigmentation	Normal
32	USH2A	c.1606 T > C;p.C536R	c. 10724G > A; p.C3575Y	6.0	Normal	Normal	Normal	Normal

Table 1 continued

Pt #	USH type	Genetic diagnosis		First ophthalmic examination				
		Mutation 1	Mutation 2	Age (years)	Disc	Vessels	Fundus	Macula
33	USH2A	c.6224G > A; p.W2075*	c.12714 T > G;p.Y4238*	8.9	Normal	Normal	Pigmentation	Altered pigment
34	USH2A	c.949C > A;p.R317R	c.(?_5299)_(5572_?)del;Exon 27 deletion	10.1	Normal	Attenuated	Normal	Normal
35	USH2A	c.2299delG; p.E767fs	c.11231 + 1G > T[§]	10.7	Normal	Attenuated	Pigmentation	Altered pigment
36	USH2A	c.2299delG; p.E767fs	c.11231 + 1G > T[§]	13.8	Normal	Attenuated	Pigmentation	altered pigment
37	USH2A	c.11864G > A; p.W3955*	c.3170A > G;p.Q1057R[§]	17.5	Normal	Attenuated	Pigmentation	NORMAL
38	USH2A	c.12295-2A > G[§]	c.1227G > C;p.W409C[§]	23.1	Normal	Attenuated	Pigmentation	Normal
USH 3								
39	CLRN1	c.189C > A;p.Y63*	c.189C > A;p.Y63*	13.3	Mild pallor	Attenuated	Pigmentation	Altered pigment

*0 = no progression, 1 = unilateral, 2 = bilateral, 4 = unknown;—= no data

[†]0 = no; 1 = yes;—= no data

[‡]Novel, not reported in the literature or ClinVar

[§]Reported in ClinVar only

[‖]Novel variant, but patient previously reported in PMID 27631835

Although ERG responses were recorded from both eyes, the b-wave amplitude from the eye with the larger (more normal) amplitude was analyzed. We used analysis of variance to compare b-wave amplitude at the first evaluation for USH1 (N = 26), USH2 (N = 11), and controls (N = 56). The amplitudes were compared to the 95% prediction limits of normal calculated from the ERG responses of 56 healthy controls [24].

Results

Of the 26 patients with USH1, 23 had pathogenic mutations in *MYO7A,* two had mutations in *CDH23*, and one had mutations in *PCDH15*. Twelve patients with USH2 had mutations in *USH2A*. In the patient with USH3, mutations were in *CLRN1*. Novel variants in *MYO7A* were identified in three patients (#2, #12, #14), and two novel variants in *CDH23* were identified in patient #26. The earliest age at which the genetic diagnosis of USH was secured is 5.4 months (#5). Four patients (#9, #19, #23, and #35) with clinical signs and symptoms of USH did not have genetic testing because they had full siblings with genetically confirmed diagnosis of USH (biallelic mutations in either *MYO7A* or *USH2A)* (Table 1). By ophthalmoscopy, fundus abnormalities were seen at the age of 5 or older in 18 of 21 patients with USH1 and USH2 (Table 1). Conversely, only four of the 18 children with USH1 or USH2 examined at less than five years had fundus abnormalities.

The majority of the cohort (26 of 39) had initial ophthalmology and otolaryngology evaluations in their first decade ($\leq$ than 10 years); for 10 patients, the initial evaluation was in infancy ($\leq$ 12 months). The ages at first visit of the patients with genetic diagnosis of USH are summarized in Table 2.

Late walking, a clinical index of vestibular dysfunction, was common in USH1 but uncommon in USH2. Independent walking at or before age 15 months is considered walking "on time" [35, 36]. Patients' otolaryngologic histories are summarized in Table 3.

Although visual acuity in the patients with USH was often below the normal mean for age, during their first decade of life the visual acuity was outside the 95% confidence limits of normal in only a few (Fig. 1). In the majority, after the first decade, acuities

Table 2 Initial assessment examinations by patients' age (years)

	Usher syndrome, clinical subtype		
	USH1	USH2	USH3
Number of patients (%)	26 (67%)	12 (31%)	1(2%)
Hearing loss detected, median age (range)	0.1 (0–2)	0 (0–4)	0.9
Otolaryngologic examination, median age (range)	0.7 (0–20)	1.1 (0–23.5)	13.0
Ophthalmic examination, median age (range)	4 (0.1–20.7)	7.5 (1.2–23.1)	13.3
ERG, median age (range)	8.1 (0.2–20.7)	9.3 (1.5–23.1)	13.3
DAT, median age (range)	8.1 (0.4–20.7)	11.6 (3.1- 23.1)	13.4
Genetic confirmation, median age (range)	6.6 (0.5–21)	9 (1.1–23.7)	13.5

were often below normal limits. With age, there was a slow but statistically significant decline of visual acuity in patients with USH1 (0.02 log unit per year; 95% CI: 0.01, 0.04; $p = 0.003$). There was no significant change of acuity with older age in patients with USH2 ($p = 0.73$).

Spherical equivalent of many with USH was within the limits of normal for age (Fig. 2). However, among those with USH1 in the first decade, hyperopia ($N = 4$) and myopia ($N = 3$) exceeded the limits of normal. Among those with USH2, spherical equivalent was within the limits of normal during the first decade.

Even at the first DAT test, among the USH groups there was significant threshold elevation compared with the *GJB2* controls ($F = 20.1$; df: 2,49; $p < 0.001$). Threshold was significantly higher in USH1 compared to USH2 [USH1: 1.55 (SD = 0.79) log unit; USH2: 0.87 (SD = 0.73) log unit]. The thresholds in all *GJB2* subjects were within the limits of normal for previously reported [22] healthy controls, as shown in Fig. 3. Patients were given

Table 3 Otolaryngologic history by clinical subtype

	Usher syndrome, clinical subtype		
	USH1 $N = 26$	USH2 $N = 12$	USH3 $N = 1$
Newborn hearing screen (NBHS)			
Pass	1	0	0
Refer*	15	5	0
Born prior to implementation of NBHS	3	3	0
No information	7	4	1
Severity of hearing loss (pure tone average of worse ear)			
Profound (> 90 dB HL)	25	1	1
Severe (71–90 dB HL)	–	–	–
Moderate–severe (56–70 dB HL)	1	7	–
Moderate (41–55 dB HL)	–	4	–
Mild (21–40 dB HL)	–	–	–
Normal (0–20 dB HL)	–	–	–
Interventions			
Bilateral cochlear implant	11	–	–
Unilateral cochlear implant	7	–	–
Bilateral hearing aide	7	12	1
Neither implant nor aide	1	–	–
Vestibular symptoms			
Age of walking median (range), months	20 (15–36)	12 (8–17)	No data

*"Refer" means infant did not pass newborn hearing screen and was referred for more definitive testing

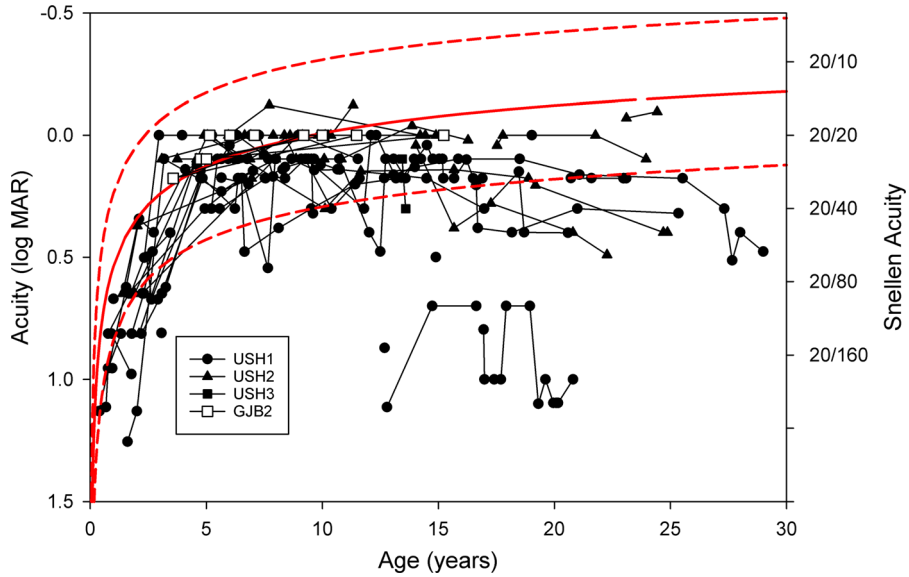


Fig. 1 Visual acuity as a function of age in patients with USH. Data from the eye with better acuity are shown. Results from an individual are connected by line segments. Results from the *GJB2* subjects are shown by opened symbols. The solid red curve represents normal mean acuity; dashed red curves represent the upper and lower 95% limits of normal [34]. The horizontal black line at log MAR 1.0 corresponds to VA = 20/200

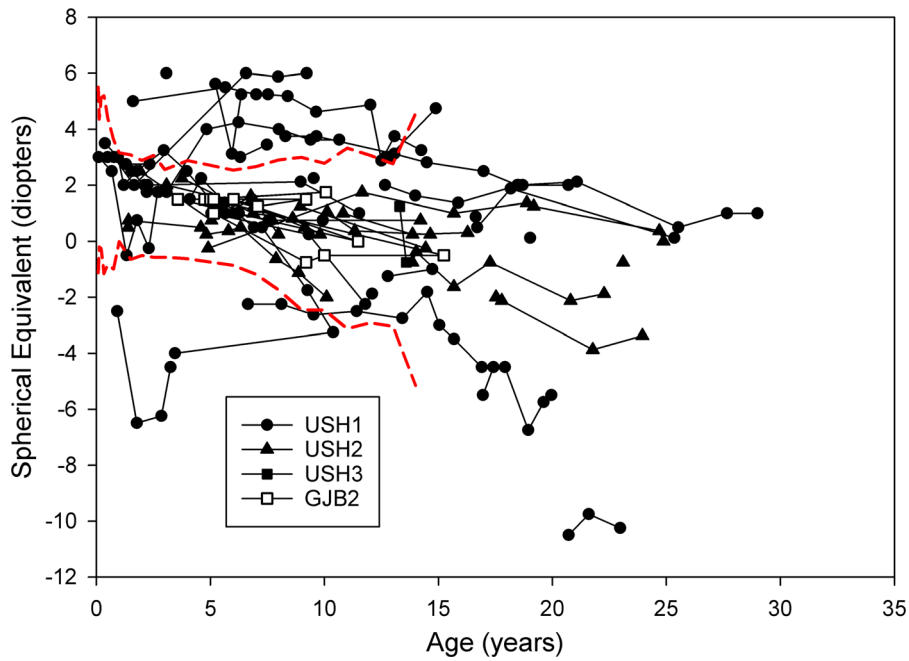


Fig. 2 Spherical equivalent as a function of age. Results from an individual are connected by line segments. Results from the *GJB2* subjects are shown by opened symbols The dashed lines indicate the upper and lower 95% prediction limits of normal acuity [32, 33]

between 1 and 10 DAT tests. At the first DAT test, all but four patients with USH1 and two with USH2 had elevated thresholds. Of the patients tested before the age of 5, thresholds were elevated in four of nine with USH1 and in all four patients with USH2. In DAT tests 2 through 10, spanning 0.6 to 19.8 (median 10) years in

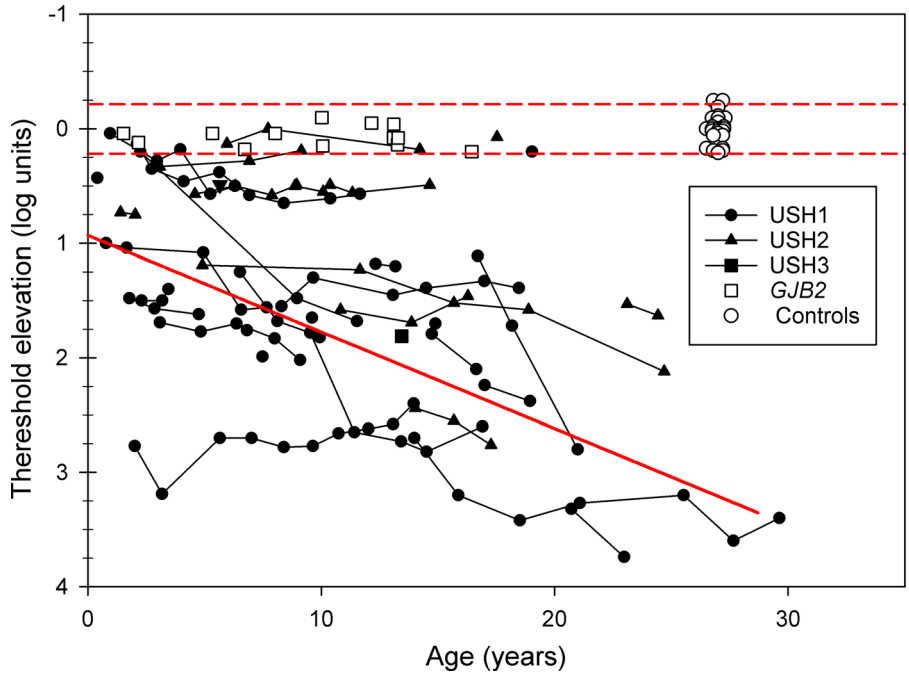


Fig. 3 Dark-adapted threshold as a function of age. The vertical axis shows threshold elevation relative to the normal mean. The horizontal solid red line at zero represents the normal mean, and the dashed lines the upper and lower 95% limits of normal. As indicated in the key, the solid symbols show DATs of individuals with USH; results from an individual are connected by line segments. Results from the *GJB2* subjects and from the young adults' controls are shown by opened symbols. The solid red line (slope 0.085 log unit per year) represents the linear mixed model fit to the USH1 thresholds and indicates increasing threshold elevation (that is, decreasing sensitivity)

23 patients with USH, longitudinal assessments showed that thresholds worsened gradually (0.085 log unit per year; 95% CI: 0.044, 0.126; $p < 0.001$) in USH1 but did not change significantly in USH2 (0.02 log unit per year; 95% CI: -0.034, 0.072; $p = 0.48$) (see Fig. 3).

Sample ERG records from patients with USH1, USH2, and a control participant are shown in Fig. 4. The amplitudes of responses are plotted as a function of age in Fig. 5. At the first ERG, measurable b-wave responses ($\geq$ 2.1 μV) were obtained in the majority of patients with USH1 (23 of 26), all patients with USH2, and the patient with USH3. For dark-adapted eyes (Fig. 5a, b), the b-wave amplitudes in all patients with USH1, 8 of 12 patients with USH2, and the patient with USH3 are below the minimum value found in any control participant (median 23 years old). In both dark-adapted conditions, b-wave amplitude differed significantly between groups ($F = 173.5$; *df*: 2.91; $p < 0.001$) ($F = 194.3$; *df*: 2.91; $p < 0.001$). Post hoc test (Tukey HSD) showed significant difference between USH1 and USH2 for Rod 0.23 condition, but not for Rod 0.01. In light-adapted conditions (Fig. 5c, d), measurable b-wave responses were obtained in 23 patients with USH1, all 12 with USH2, and the 1 with USH3; b-wave amplitude was below the control minimum in all USH1, three USH2, and the patient with USH3 (Fig. 5c). Responses to the 30 Hz flicker stimulus were recorded in 24 of the 26 USH1; all were below the control minimum. Responses to the 30 Hz flicker stimulus in 5 of 11 with USH2 and the patient with USH3 were below the control minimum (Fig. 5d). In both light-adapted conditions, b-wave amplitude differed significantly between groups ($F = 109.8$; *df*: 2.83; $p < 0.001$); ($F = 75.4$; *df*: 2,72; $p < 0.001$). Post hoc test (Tukey HSD) showed the USH groups differed significantly from controls and from each other, with USH1 responses more attenuated than USH2. Implicit time, in both dark- and light-adapted conditions, was longer than the normal range in the majority of patients with USH.

All nine with USH1 who were tested prior to age five years had abnormal ERG responses in both dark-

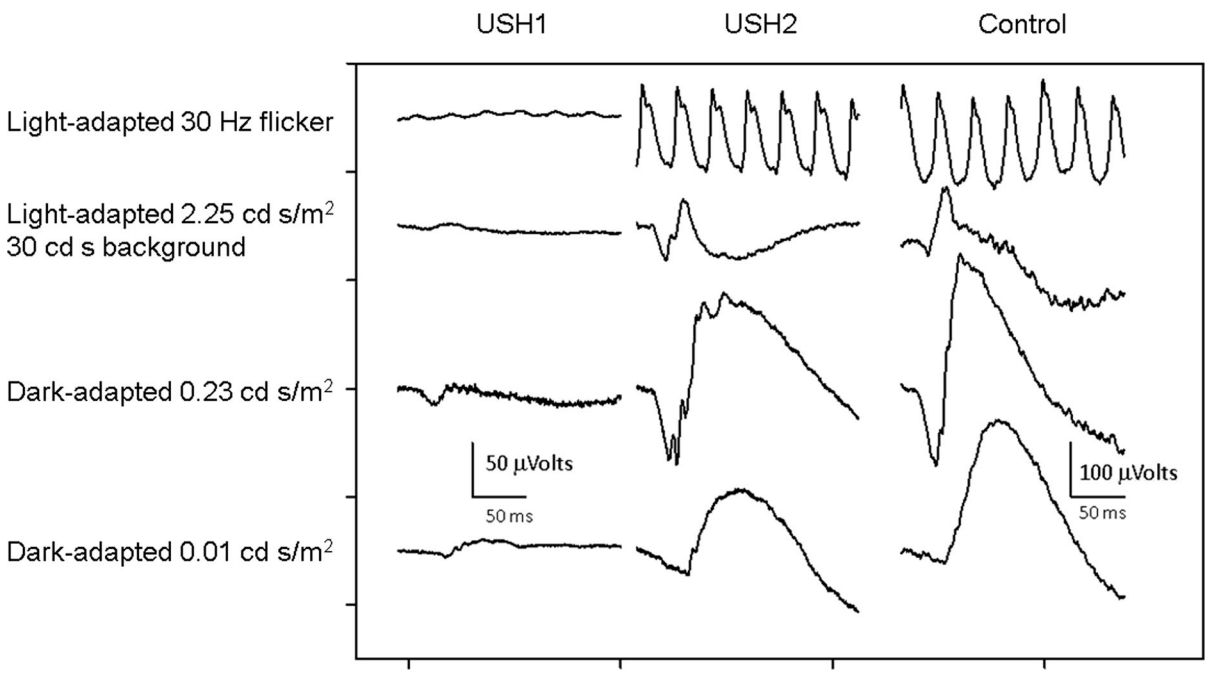


Fig. 4 Sample ERG waveforms by comparison groups. Sample ERG records from a patient with USH1 (#6), a patient with USH2 (#33), and a healthy control. Stimulus conditions are shown on the left. Note that the calibration bars for USH patients are shown on the left and for the control patient on the right. Each trace starts at flash (less than 1 ms duration) onset

and light-adapted conditions. The three patients with USH2 tested prior to age five had b-wave amplitudes within the normal range in both dark- and light-adapted conditions. If we consider ERG tests done by age 10 years, all USH1 patients had responses below normal, but many with USH2 had normal amplitudes (Fig. 5a–d). The median age at initial ERG was < 10 years (8 years in USH1 or 9 years in USH2) (Table 2).

Of the 24 patients with USH1 who had both DAT and ERG, 20 had both abnormal ERG and DAT at the first examination (median age 8 years). Four patients had normal DATs and abnormal ERGs; in other words, the DAT test would have missed diagnosis of disease in those patients. None had both normal DAT and ERG.

Of the 10 patients with USH2 who had both DAT and ERG, seven had both abnormal tests at the first examination (median age 9 years). Only one patient had a normal DAT and abnormal ERG; two patients had abnormal DATs and normal ERGs. None had both normal DAT and ERG.

Discussion

Among our patients with secured genetic diagnoses of USH, the results of initial evaluation of retinal function, including DAT and full-field ERG, were abnormal in the majority of patients (USH1, 83%; USH2, 70%). At initial test, the ERG was more frequently abnormal than the DAT. At the initial test, we never found both normal DAT *and* normal ERG in any patient. Despite the strong association of abnormal retinal function and the diagnosis of USH, some young patients would not have been detected by retinal function testing alone. Specifically, 17% of USH1 and 30% of USH2 would have been missed by the initial tests of retinal function (ERG + DAT) performed at median age < 10 years (Table 1). Although we have taken genetic diagnosis as the gold standard for a secure diagnosis of USH, we are reminded that the genetic diagnosis does not assess severity of retinal disorder nor does it necessarily forecast the course of visual impairment.

The course of visual acuity (Fig. 1) documented deficits in the first decade and significant decline thereafter in USH1 but not in USH2. However, the

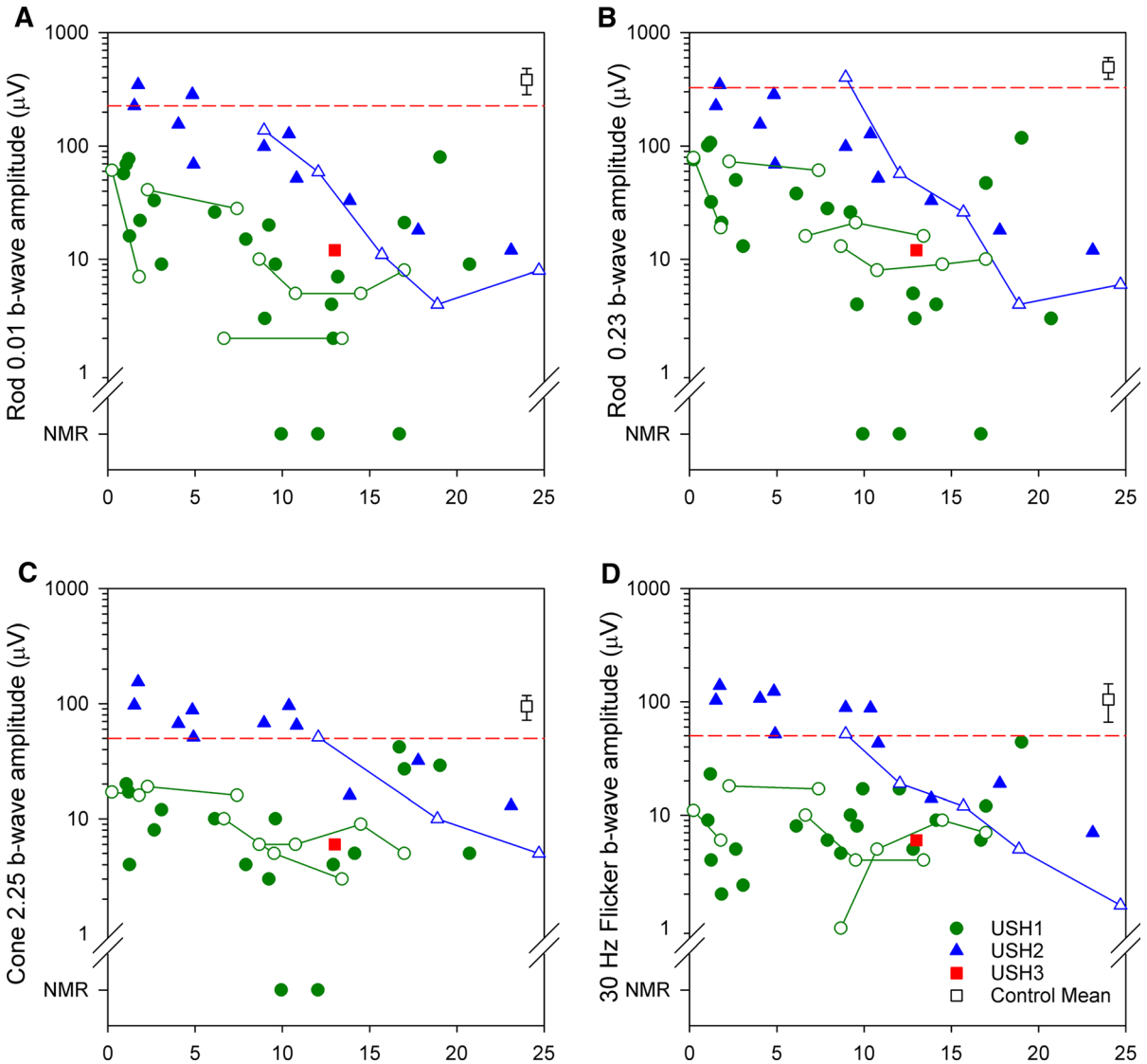


Fig. 5 ERG b-wave amplitude as a function of age. Sequential measures (open symbols) from an individual are connected by line segments. The normal mean is indicated by the open square; error bars represent one standard deviation. The horizontal dashed line represents the lower 95% limit of normal. NMR indicates no measurable response (that is, less than 2.1 μV)

data for USH2 were quite sparse. Less than 20% of all of our patients with USH had spherical equivalent that exceeded the limits of normal. Thus, high refractive error is unlikely to be a diagnostic clue. Fundus abnormalities were quite uncommon in our patients younger than 5 years but frequent in older children (Table 1). Thus, clinical assessment in the absence of tests of retinal function would have done little to detect USH in these pediatric patients. We did note progressive myopia in both USH1 and USH2; this underscores the importance of ongoing ophthalmic care and reminds us that annual ophthalmic examination is advised for children with hearing loss [34, 35].

The specific diagnosis of USH depends on confirmatory genetic testing. For one thing, genetic testing is essential for distinguishing USH at an early age from other causes of hearing loss plus retinal disease (*ALMS1, ABHD12, TUBB4B*, *SLC52A2*, and *CHD7)* in an individual [37–40]. Fortunately, since our first observations (1994) there have been enormous advances in identifying additional USH genes and in the increased availability of testing for these genes.

Despite the implementation of Universal Newborn Hearing Screening with subsequent earlier identification of hearing loss and improved genetic testing, there is often still a lag in securing a specific molecular diagnosis. Continued barriers to obtaining necessary genetic testing include insurance pushback, family apprehension, and cost. Nevertheless, given that USH presents as non-syndromic hearing loss in infants and young children, clinicians must suspect Usher in young children or no genetic testing would be ordered at that time. Although conducting both DAT and ERG may capture a larger group of patients with USH at earlier ages, these are not routinely performed or even routinely available tests in young children and may often only be requested if clinical vision is failing, or if a molecular diagnosis has already been secured.

Conclusion

Although Usher Syndrome is the most common genetic form of deaf-blindness, it is still considered a rare disease. Diagnosis depends on initial clinical suspicion, accurate genetic testing, and ophthalmological findings. A routine ophthalmology examination will not diagnose (or rule out) USH in infants and young children, and even ERG and DAT in young children may be within the wide range of normal in children with genetically confirmed USH. Given that ophthalmological abnormalities are common in infants and children with SNHL, genetic testing is the most accurate strategy to confirm a diagnosis of USH. As genetic therapies become available for USH patients, it will be increasingly important to diagnosis these patients early, when retinal function and useful vision remain.

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Ethics approval The study conformed to the tenets of the Declaration of Helsinki and was approved by the Children's Hospital Committee on Clinical Investigation.

Statement of human rights All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards.

Statement on the welfare of animals This article does not contain any studies with animals performed by any of the authors.

Informed consent Informed consent was obtained from a individual participants included in the study.

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